HW3

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load library

```
library(haven)
```

```
Warning: package 'haven' was built under R version 4.5.3
```

```
library(knitr)
```

```
Warning: package 'knitr' was built under R version 4.5.3
```

Problem 1

a. Download and Read File

```
download.file("https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2015/DataFiles/AUX_I.xpt", "AUX_
download.file("https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2015/DataFiles/DEMO_I.xpt", "DEM

aux <- read_xpt("AUX_I.XPT")
demo <- read_xpt("DEMO_I.XPT")

dim(aux)
```

```
[1] 4582   73
```

```
dim(demo)
```

```
[1] 9971   47
```

```
names(aux)
```

```
 [1] "SEQN"     "AUAEXSTS" "AUAEXCMT" "AUQ010"   "AUQ020"   "AUQ020A"
 [7] "AUQ020B"  "AUQ020C"  "AUQ020D"  "AUQ020E"  "AUQ030"   "AUQ040"
[13] "AUQ050"   "AUXOTSPL" "AUXLOEXC" "AUXLOIMC" "AUXLOCOL" "AUXLOABN"
[19] "AUDLOABC" "AUXROTSP" "AUXROEXC" "AUXROIMC" "AUXROCOL" "AUXROABN"
[25] "AUDROABC" "AUXTMEPR" "AUXTPVR"  "AUXTWIDR" "AUXTCOMR" "AUXTMEPL"
[31] "AUXTPVL"  "AUXTWIDL" "AUXTCOML" "AUAEAR"   "AUAMODE"  "AUAFMANL"
[37] "AUAFMANR" "AUXU1K1R" "AUXU500R" "AUXU1K2R" "AUXU2KR"  "AUXU3KR"
[43] "AUXU4KR"  "AUXU6KR"  "AUXU8KR"  "AUXU1K1L" "AUXU500L" "AUXU1K2L"
[49] "AUXU2KL"  "AUXU3KL"  "AUXU4KL"  "AUXU6KL"  "AUXU8KL"  "AUXR1K1R"
[55] "AUXR5CR"  "AUXR1K2R" "AUXR2KR"  "AUXR3KR"  "AUXR4KR"  "AUXR6KR"
[61] "AUXR8KR"  "AUXR1K1L" "AUXR5CL"  "AUXR1K2L" "AUXR2KL"  "AUXR3KL"
[67] "AUXR4KL"  "AUXR6KL"  "AUXR8KL"  "AUAREQC"  "AUATYMTR" "AUALEQC"
[73] "AUATYMTL"
```

```
names(demo)
```

```
 [1] "SEQN"     "SDDSRVYR" "RIDSTATR" "RIAGENDR" "RIDAGEYR" "RIDAGEMN"
 [7] "RIDRETH1" "RIDRETH3" "RIDEXMON" "RIDEXAGM" "DMQMILIZ" "DMQADFC"
[13] "DMDBORN4" "DMDCITZN" "DMDYRSUS" "DMDEDUC3" "DMDEDUC2" "DMDMARTL"
[19] "RIDEXPRG" "SIALANG"  "SIAPROXY" "SIAINTRP" "FIALANG"  "FIAPROXY"
[25] "FIAINTRP" "MIALANG"  "MIAPROXY" "MIAINTRP" "AIALANGA" "DMDHHSIZ"
[31] "DMDFMSIZ" "DMDHHSZA" "DMDHHSZB" "DMDHHSZE" "DMDHRGND" "DMDHRAGE"
[37] "DMDHRBR4" "DMDHREDU" "DMDHRMAR" "DMDHSEDU" "WTINT2YR" "WTMEC2YR"
[43] "SDMVPSU"  "SDMVSTRA" "INDHHIN2" "INDFMIN2" "INDFMPIR"
```

merge into a dataframe

AUXTWIDR Tympanometric width, right ear AUXTWIDL Tympanometric width, left ear

```
attr(aux$AUXTWIDR, "label")
```

```
[1] "Tympanometric width, right ear"
```

```
attr(aux$AUXTWIDL, "label")
```

```
[1] "Tympanometric width, left ear"
```

```r
# check these labels are about Tympanometric
summary(aux$AUXTWIDR)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   8.00   61.00   79.00   84.73   98.00  479.00     433
```

```r
summary(aux$AUXTWIDL)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   2.00   62.00   78.00   85.34   98.00  479.00     479
```

```r
# check the number of NA's
```

```r
aux_s <- aux[, c("SEQN", "AUXTWIDR", "AUXTWIDL")]
names(aux_s) <- c("SEQN", "tymp_r", "tymp_l")

demo_s <- demo[, c("SEQN", "RIAGENDR", "DMDCITZN", "DMDHHSZA", "INDHHIN2")]
names(demo_s) <- c("SEQN", "gender_raw", "citizen_raw", "children_raw", "income_raw")
# get data with only labels we need for AUX and demo
```

```r
data <- merge(
  x = aux_s,
  y = demo_s,
  by = "SEQN",
  all = FALSE
)
# merge AUX and demo
```

```r
dim(data)
```

```
[1] 4582    7
```

```r
# Display the dimensions of the data.frame
```

b. Clean the Variables

```r
table(data$gender_raw, useNA = "always")
```

```
   1    2 <NA>
2176 2406    0
```

```r
table(data$citizen_raw, useNA = "always")
```

```
   1    2    7    9 <NA>
3684  884    8    5    1
```

```r
table(data$children_raw, useNA = "always")
```

```
   0    1    2    3 <NA>
3463  737  317   65    0
```

```r
table(data$income_raw, useNA = "always")
```

```
   1    2    3    4    5    6    7    8    9   10   12   13   14   15   77   99
 101  146  233  262  272  461  439  381  295  244  145   53  458  795   86   63
<NA>
 148
```

```r
summary(data$tymp_r)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   8.00   61.00   79.00   84.73   98.00  479.00     433
```

```r
summary(data$tymp_l)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   2.00   62.00   78.00   85.34   98.00  479.00     479
```

```r
# check if there are any weird numbers like 999
```

```r
data$tymp_r <- as.numeric(data$tymp_r)
data$tymp_l <- as.numeric(data$tymp_l)

data$tymp_r[data$tymp_r %in% c(555, 777, 999)] <- NA
data$tymp_l[data$tymp_l %in% c(555, 777, 999)] <- NA
# recode special numbers as NA

data$tymp_r[!is.na(data$tymp_r) & (data$tymp_r < 0 | data$tymp_r > 499)] <- NA
data$tymp_l[!is.na(data$tymp_l) & (data$tymp_l < 0 | data$tymp_l > 499)] <- NA
# the value must be in the range of 1 ~ 499

summary(data$tymp_r)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   8.00   61.00   79.00   84.73   98.00  479.00     433
```

```r
summary(data$tymp_l)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   2.00   62.00   78.00   85.34   98.00  479.00     479
```

```r
# we check the summary again but we can see that there are no invalid value
```

```r
data$gender_raw[data$gender_raw %in% c(7, 9)] <- NA
data$gender <- factor(
  data$gender_raw,
  levels = c(1, 2),
  labels = c("male", "female")
)
# convert gender to a factor with informative labels.
# male is the first level

data$citizen_raw[data$citizen_raw %in% c(7, 9)] <- NA
data$citizen <- factor(
  data$citizen_raw,
  levels = c(1, 2),
  labels = c("citizen", "noncitizen")
)
```

```r
# convert citizenship status to a factor with informative labels.

data$children_raw[data$children_raw %in% c(7, 9)] <- NA
data$children <- as.numeric(data$children_raw)
data$citizen[!(data$children %in% 0: 3)] <- NA
# convert the number of young children to an ordinary numeric variable.

data$income_raw[data$income_raw %in% c(12, 13, 77, 99)] <- NA
income_midpoint <- c(
  "1"  = 2500,
  "2"  = 7500,
  "3"  = 12500,
  "4"  = 17500,
  "5"  = 22500,
  "6"  = 30000,
  "7"  = 40000,
  "8"  = 50000,
  "9"  = 60000,
  "10" = 70000,
  "14" = 87500,
  "15" = 100000
)
data$income <- unname(income_midpoint[as.character(data$income_raw)])
# convert detailed income categories to approximate dollar midpoints.
# codes 12 and 13 are broad categories that cannot be mapped uniquely.
# codes 77 and 99 are documented missing-value codes.
# all four codes therefore become NA in the continuous variable.

# replace special value in all variables with NA

table(data$gender, useNA = "always")
```

```
  male female   <NA>
  2176   2406      0
```

```r
table(data$citizen, useNA = "always")
```

```
   citizen noncitizen       <NA>
      3684        884         14
```

```r
table(data$children, useNA = "always")
```

```
   0    1    2    3 <NA>
3463  737  317   65    0
```

```r
table(data$income, useNA = "always")
```

```
 2500  7500 12500 17500 22500 30000 40000 50000 60000 70000 87500 1e+05  <NA>
  101   146   233   262   272   461   439   381   295   244   458   795   495
```

c. Fit Poisson Model

```r
modelrp1 <- glm(
  tymp_r ~ gender,
  data = data,
  family = poisson(),
  na.action = na.omit
)
modelrp2 <- glm(
  tymp_r ~ gender + citizen + children + income,
  data = data,
  family = poisson(),
  na.action = na.omit
)
modellp1 <- glm(
  tymp_l ~ gender,
  data = data,
  family = poisson(),
  na.action = na.omit
)
modellp2 <- glm(
  tymp_l ~ gender + citizen + children + income,
  data = data,
  family = poisson(),
  na.action = na.omit
)
# construct four Poisson models
```

```r
summary(modelrp1)
```

```
Call:
glm(formula = tymp_r ~ gender, family = poisson(), data = data,
    na.action = na.omit)

Coefficients:
             Estimate Std. Error  z value Pr(>|z|)
(Intercept)  4.434588   0.002468 1796.797  < 2e-16 ***
genderfemale 0.009264   0.003380    2.741  0.00613 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 70970  on 4148  degrees of freedom
Residual deviance: 70963  on 4147  degrees of freedom
  (433 observations deleted due to missingness)
AIC: 96618

Number of Fisher Scoring iterations: 4
```

```r
summary(modelrp2)
```

```
Call:
glm(formula = tymp_r ~ gender + citizen + children + income,
    family = poisson(), data = data, na.action = na.omit)

Coefficients:
                    Estimate Std. Error  z value Pr(>|z|)
(Intercept)        4.469e+00  4.332e-03 1031.781  < 2e-16 ***
genderfemale       1.456e-02  3.595e-03    4.050 5.13e-05 ***
citizennoncitizen  4.437e-02  4.613e-03    9.618  < 2e-16 ***
children          -5.627e-03  2.713e-03   -2.074   0.0381 *
income            -9.062e-07  5.646e-08  -16.051  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 62369  on 3704  degrees of freedom
Residual deviance: 61954  on 3700  degrees of freedom
  (877 observations deleted due to missingness)
AIC: 84858

Number of Fisher Scoring iterations: 4
```

```
summary(modellp1)
```

```
Call:
glm(formula = tymp_l ~ gender, family = poisson(), data = data,
    na.action = na.omit)

Coefficients:
             Estimate Std. Error  z value Pr(>|z|)
(Intercept)  4.439805   0.002473 1795.263  < 2e-16 ***
genderfemale 0.012952   0.003387    3.824 0.000131 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 73307  on 4102  degrees of freedom
Residual deviance: 73293  on 4101  degrees of freedom
  (479 observations deleted due to missingness)
AIC: 98685

Number of Fisher Scoring iterations: 4
```

```
summary(modellp2)
```

```
Call:
glm(formula = tymp_l ~ gender + citizen + children + income,
    family = poisson(), data = data, na.action = na.omit)

Coefficients:
                    Estimate Std. Error  z value Pr(>|z|)
```

```
(Intercept)        4.469e+00  4.346e-03 1028.355  < 2e-16 ***
genderfemale       1.863e-02  3.600e-03    5.176 2.26e-07 ***
citizennoncitizen  2.151e-02  4.648e-03    4.627 3.71e-06 ***
children          -1.707e-02  2.728e-03   -6.257 3.93e-10 ***
income            -6.386e-07  5.637e-08  -11.330  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 64149  on 3664  degrees of freedom
Residual deviance: 63925  on 3660  degrees of freedom
  (917 observations deleted due to missingness)
AIC: 86607

Number of Fisher Scoring iterations: 4
```

produce a table

```r
irr <- function(model, model_name){
  coef_table <- summary(model)$coefficients
  beta <- coef_table[, "Estimate"]
  se <- coef_table[, "Std. Error"]
  p_value <- coef_table[, "Pr(>|z|)"]
  data.frame(
    model = model_name,
    predictor = rownames(coef_table),
    IRR = exp(beta),
    CI_lower = exp(beta - 1.96 * se),
    CI_upper = exp(beta + 1.96 * se),
    p_value = p_value,
    row.names = NULL
  )
}

irr_result <- rbind(
  irr(modelrp1, "1R - Right ear: gender"),
  irr(modelrp2, "2R - Right ear: gender, citizenship status, number of children, annual house
  irr(modellp1, "1L - Left ear: gender"),
  irr(modellp2, "2L - Right ear: gender, citizenship status, number of children, annual house
)
irr_result
```

	model	predictor	IRR	CI_lower	CI_upper	p_value
1	1R - Right ear: gender	(Intercept)	84.3174114	83.9105209	84.7262750	0.000000e+00
2	1R - Right ear: gender	genderfemale	1.0093073	1.0026420	1.0160169	6.134056e-03
3	2R - Right ear: gender, citizenship status, number of children, annual household income	(Intercept)	87.3008886	86.5628307	88.0452394	0.000000e+00
4	2R - Right ear: gender, citizenship status, number of children, annual household income	genderfemale	1.0146661	1.0075414	1.0218412	5.126244e-05
5	2R - Right ear: gender, citizenship status, number of children, annual household income	citizennoncitizen	1.0453685	1.0359587	1.0548638	6.739069e-22
6	2R - Right ear: gender, citizenship status, number of children, annual household income	children	0.9943886	0.9891141	0.9996911	3.809399e-02
7	2R - Right ear: gender, citizenship status, number of children, annual household income	income	0.9999991	0.9999990	0.9999992	5.586202e-58
8	1L - Left ear: gender	(Intercept)	84.7584241	84.3485762	85.1702635	0.000000e+00
9	1L - Left ear: gender	genderfemale	1.0130366	1.0063334	1.0197844	1.313548e-04
10	2L - Right ear: gender, citizenship status, number of children, annual household income	(Intercept)	87.3120509	86.5714291	88.0590088	0.000000e+00
11	2L - Right ear: gender, citizenship status, number of children, annual household income	genderfemale	1.0188077	1.0116451	1.0260210	2.261460e-07
12	2L - Right ear: gender, citizenship status, number of children, annual household income	citizennoncitizen	1.0217415	1.0124750	1.0310928	3.707248e-06
13	2L - Right ear: gender, citizenship status, number of children, annual household income	children	0.9830788	0.9778372	0.9883486	3.931980e-10
14	2L - Right ear: gender, citizenship status, number of children, annual household income	income	0.9999994	0.9999993	0.9999995	9.284589e-30